

Propositional Logic II

Previous Lecture

● *Propositions, primitive and compound*

- p where p is "it is not raining"
 - $\neg p$ where p is "it is raining"
- Same meaning*

● *Logic connectives:*

- *negation* $\neg$
- *conjunction* $\wedge$
- *disjunction* $\vee$
- *exclusive or* $\oplus$
- *implication* $\rightarrow$
- *biconditional* $\leftrightarrow$

● *Truth tables*

Example: converting English to logic

'You can access the Internet from campus if you are a computer science major or if you are not a freshman.'

p - *'you can access the Internet from campus'*

q - *'you are a computer science major'*

r - *'you are a freshman'*

Nested formulas

- Can also combine *compound propositions* with logical connectives

e.g. $(p \rightarrow q) \rightarrow p$

- Truth value depends on the truth value of the *sub-formulas*

p	q	$p \rightarrow q$	$(p \rightarrow q) \rightarrow p$
0	0	1	
0	1	1	
1	0	0	
1	1	1	

An Example

- *Construct the truth table of the following compound proposition*

$$p \rightarrow (\neg q \vee p)$$

p	q	$\neg q \vee p$	$p \rightarrow (\neg q \vee p)$
0	0		
0	1		
1	0		
1	1		

Tautologies

- A *Tautology* is a compound proposition (formula) that is *true* for all combinations of truth values of its propositional variables

$$(p \rightarrow q) \vee (q \rightarrow p)$$

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \vee (q \rightarrow p)$
0	0			
0	1			
1	0			
1	1			

“To be or not to be”

Contradictions

- **Contradiction** is a compound proposition (formula) that is **false** for all combinations of truth values of its propositional variables

$$(p \oplus q) \wedge (p \oplus \neg q)$$

p	q	$p \oplus q$	$p \oplus \neg q$	$(p \oplus q) \wedge (p \oplus \neg q)$
0	0			
0	1			
1	0			
1	1			

“Black is white and black is not white”

Example



Write the following as a propositional formula and construct the truth table

“An inhabitant of a castle in Transylvania is either sane or insane, and is a human or a vampire”



Example

Write the following as a propositional formula and construct the truth table

“If a person is an insane vampire then he believes only in false things and always lies”

Logic Equivalences

- Compound propositions Φ and Ψ are said to be **logically equivalent** if the proposition Φ is true (false) if and only if Ψ is true (respectively, false)

or

- The truth tables of Φ and Ψ are equal

or

- For any choice of truth values of the primitive propositions (propositional variables) of Φ and Ψ , formulas Φ and Ψ have the same truth value

- If Φ and Ψ are logically equivalent, we write

$$\Phi \Leftrightarrow \Psi$$

Why Logic Equivalences

- *To simplify compound propositions*

“If you are a computer science major or a freshman and you are not a computer science major or you are granted access to the Internet, then you are a freshman or have access to the Internet”

- *To convert complicated compound propositions to certain ‘normal form’ that is easier to handle*

Conjunctive Normal Form CNF

Example Equivalences

● *Implication and its contrapositive*

p	q	$p \rightarrow q$	$\neg q \rightarrow \neg p$
0	0	1	
0	1	1	
1	0	0	
1	1	1	

● *All tautologies are equivalent to T*

● *All contradictions are equivalent to F*

Equivalences and Tautologies

- **Theorem** *Compound propositions Φ and Ψ are logically equivalent **if and only if** $\Phi \leftrightarrow \Psi$ is a tautology.*

*To **prove** the above theorem, need to show two things:*

- *“if” direction:*
if $\Phi \leftrightarrow \Psi$ is a tautology, then $\Phi \Leftrightarrow \Psi$
- *“only if” direction:*
if $\Phi \Leftrightarrow \Psi$, then $\Phi \leftrightarrow \Psi$ is a tautology

Equivalences and Tautologies cont.

● **Theorem** *Compound propositions Φ and Ψ are logically equivalent **if and only if** $\Phi \leftrightarrow \Psi$ is a tautology.*

Proof (“if” direction):

- **Assume** $\Phi \leftrightarrow \Psi$ is a tautology.
- Then **by definition of a tautology**, $\Phi \leftrightarrow \Psi$ is True for every choice of truth values for primitive propositions.
- Since $\Phi \leftrightarrow \Psi$ is True if and only if Φ and Ψ have the same truth value, for every choice of truth values of propositional variables, Φ and Ψ have the same truth value.
- Hence by definition, $\Phi \leftrightarrow \Psi$

Equivalences and Tautologies cont.

● **Theorem** *Compound propositions Φ and Ψ are logically equivalent if and **only if** $\Phi \leftrightarrow \Psi$ is a tautology.*

Proof (“only if” direction):

- **Assume** $\Phi \leftrightarrow \Psi$ is a tautology.
- Then **by definition of** $\Phi \leftrightarrow \Psi$, for every choice of truth values of propositional variables, Φ is True if and only if Ψ is True.
- Then $\Phi \leftrightarrow \Psi$ is True for every choice of truth values of propositional variables.
- Hence by definition, $\Phi \leftrightarrow \Psi$ is a Tautology.

Q.E.D.

Practice

Exercises from the Book:

7th edition: No. 10(c,d), 14, 16, 17, 22, 31 (page 35)

8th edition: No. 12(c,d), 18, 20, 21, 26, 35 (page 38)

Practice

- *Construct a truth table*

$$(p \rightarrow q) \vee (\neg p \rightarrow r)$$

- *Are these system specifications consistent (not a contradiction)?*
 - *The router can send packets to the edge system only if it supports the new address space*
 - *For the router to support the new address space it is necessary that the latest software release be installed.*
 - *The router can send packets to the edge system if the latest software release is installed.*
 - *The router does not support the new address space.*